

24th Conference on Applications of Computer Algebra

Santiago de Compostela. June 18-22, 2018

Plenary Session

Aula Magna	title	Invited speaker
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Tuesday 19th

9:30 - 10:30	Dealing with real algebraic curves and surfaces for discovery: from experiments to theory and applications	Laureano González-Vega
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Wednesday 20th

9:00 - 10:00	Applications of Computer Algebra to Verification and Satisfiability Checking	James H. Davenport
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Thursday 21st

9:00 - 10:00	SAT Solvers and Computer Algebra Systems: A Powerful Combination for Mathematics	Vijay Ganesh
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Friday 22nd

9:30 - 10:30	Automatic Geometric Theorem Proving and Discovering Using (Comprehensive) Groebner Bases	Dingkang Wang
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S 01 General Session

Aula 1	title	speaker
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Thursday 21st

16:30 - 17:00	Randomized Algorithms for Normal Basis in Characteristic Zero	Armin Jamshidpey
<i>coffee break</i>		
17:30 - 18:00	Computer Algebra and Computer Science	Gereon Kremer
18:00 - 18:30	Conversion of element representations in Galois rings	Juan Carlos Ku-Cauch
18:30 - 19:00	Automatic generation of diagrammatic subway maps for any date with Maple	Eugenio Roanes-Lozano
19:00 - 19:30	Detecting truth, just on parts, in automated reasoning in geometry	Pilar Vélez

S 02 Computer Algebra Modeling in Science and Engineering

Aula 7	title	speaker
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Thursday 21st

10:00 - 10:30	Leap-Frog Algorithm for interpolating reduced sparse data	Ryszard Kozera
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10:30 - 11:00	Reparameterization and piecewise cubics for interpolating reduced data	Ryszard Kozera
<i>coffee break</i>		
11:30 - 12:00	Visualization of Planetary Motions Using KeTCindy	Satoshi Yamashita
12:00 - 12:30	Computing Perturbations in Two-Planetary Three-Body Problem with Masses Varying Non-Isotropically at Different Rates	Mukhtar Minglibayev
12:30 - 13:00	Motion of two bodies coupled by a spring on a rough plane with variable coefficient of friction: simulation with Mathematica	Alexander N. Prokopenya
<i>lunch</i>		
16:30 - 17:00	A study of sensitivity of nonlinear oscillations of a CLD parallel circuit to parametrization of Esaki diode	Haiduke Sarafian
<i>coffee break</i>		
17:30 - 18:00	Numerical study of multiphase flow and viscous fingering in a heterogeneous porous medium	Hassane Djebouri
18:00 - 18:30	3D Stress analysis of a loaded birefringent sphere by photoelastic experiment and finite elements method	Kamel Touahir
18:30 - 19:00	Fluid/Particles Flow Simulation by Finite Volume Method —Hybrid Approach—	Salah Zouaqui

S 03 Computer Algebra in Education

Aula 8	title	speaker
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Tuesday 19th

10:30 - 11:00	About the Bulgarian experience in organizing National Student Olympiad in Computer Mathematics	Penka Georgieva
<i>coffee break</i>		
11:30 - 12:00	Student Attitudes toward Technology Use in Math Education	Karsten Schmidt
12:00 - 12:30	Technology enhanced e-assessments in Calculus courses with application of CAS	Elena Varbanova
12:30 - 13:00	Analyzing the "Calculator Effect" of Different Kinds of Software for School Arithmetics and Algebra	Rein Prank
13:00 - 13:30	Dynamic visualizations for network flow optimizations problems with Mathematica	Włodzimierz Wojas - Jan Krupa
<i>lunch</i>		
15:30 - 16:00	Using TI-Nspire for the financial education of future engineers	Hanan Smidi
16:00 - 16:30	Accurate plotting in 3D: how to choose the mesh	David G. Zeitoun
16:30 - 17:00	Addressing discrete mathematics problems in the classroom	Anouk Bergeron-Brlek
<i>coffee break</i>		
17:30 - 18:00	Introducing parametric curves with CAS	Louis-Xavier Proulx

18:00 - 18:30	New rules for improving Cas capabilities when computing improper integrals. Applications in Math Education	José Luis Galán-García
18:30 - 19:00	Teaching Partial Differential Equations with CAS	José Luis Galán-García

Wednesday 20th

10:00 - 10:30	Do we take advantage of ICT when teaching maths at primary and secondary education levels? Do we teach maths as we should?	Eugenio-Roanes Lozano
10:30 - 11:00	Visualizations of the nondominated set and the efficient set in multicriteria optimization problems using Mathematica	Włodzimierz Wojas - Jan Krupa
<i>coffee break</i>		
11:30 - 12:00	Analyzing discrete suspended chains using computer algebra	Gilbert Labelle
12:00 - 12:30	Fractals and tessellations: from K's to cosmology	Thierry Dana-Picard
12:30 - 13:00	Periodic and Nontrivial Periodic Input in Linear ODEs (Part I)	Michel Beaudin
13:00 - 13:30	Periodic and Nontrivial Periodic Input in Linear ODEs (Part II)	Michel Beaudin

Thursday 21st

10:00 - 10:30	Consolidation of abstract knowledge in the process of confronting errors using digital tools: The case of the inflection point	Anatoli Kouropatov
10:30 - 11:00	The Runge Example for Interpolation and Wilkinson's Examples for Rootfinding	Leili Rafiee Sevyeri
<i>coffee break</i>		
11:30 - 12:00	A non-iterative method for solving nonlinear equations	Michael Xue
12:00 - 12:30	What is the integral of x^n ?	David J. Jeffrey
12:30 - 13:00	CAS in Teaching Basics of Stereoscopy	Benjamin Jurell
<i>lunch</i>		
16:30 - 17:00	Familiarizing students with definition of Lebesgue measure using Mathematica - some examples of calculation directly from its definition	Włodzimierz Wojas - Jan Krupa

S 04 Applied and Computational Algebraic Topology

Aula 9	title	speaker
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Tuesday 19th

10:30 - 11:00	Reductions of monomial resolutions for the computation of high dimensional simplicial homology	Eduardo Sáenz-de-Cabezón
<i>coffee break</i>		
11:30 - 12:00	New algorithms for computing homology of finite topological spaces	Julián Cuevas-Rozo
12:00 - 12:30	Computing Homotopy Information of 4D Digital Objects in Parallel	Pedro Real

12:30 - 13:00	Maximal Stable Homological Regions and AT-models	Pedro Real
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S 05 Computer Algebra for Dynamical Systems and Celestial Mechanics

Aula 10	title	speaker
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Tuesday 19th

10:30 - 11:00	On the Numerical Analysis and Visualisation of Implicit Ordinary Differential Equations	Werner M. Seiler
<i>coffee break</i>		
11:30 - 12:00	Singular Initial Value Problems for Quasi-Linear Ordinary Differential Equations	Werner M. Seiler
12:00 - 12:30	The construction of averaged semi-analytical planetary motion theory up to the third degree of planetary masses by means CAS Piranha	Alexander Perminov
12:30 - 13:00	Local and Global Properties of ODEs	Victor Edneral
<i>lunch</i>		
15:30 - 16:00	Nonlinear Oscillations of a Spring Pendulum at the 1 : 1 : 2 Resonance by Normal Form Method	Victor Edneral
16:00 - 16:30	Schutzenberger transformation on the three-dimensional Young graph	Vasilii Duzhin
16:30 - 17:00	On the estimation of complexity of trajectories in the equal-mass free-fall three-body problem	Aleksandr Mylläri
<i>coffee break</i>		
17:30 - 18:00	The modeling of the effect of velocity of breakup in osculating orbital elements of the young asteroid family	Alexey Rosaev
18:00 - 18:30	Searching for periodic solutions with central symmetry in Hill problem	Alexander Batkhin

S 06 Computational Differential and Difference Algebra

Aula 9	title	speaker
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Thursday 21st

11:30 - 12:00	Bounds for Proto-Galois Groups	Eli Amzallag
12:00 - 12:30	The global dimension of the algebras of integro-differential operators and their factor algebras	V. V. Bavula
12:30 - 13:00	Effective calculation in studying the Jacobian Conjecture	Paweł Bogdan
<i>lunch</i>		
16:30 - 17:00	Dimension Polynomials and the Einstein's Strength of Some Systems of Quasi-linear Algebraic	Alexander Levin
<i>coffee break</i>		
17:30 - 18:00	Formal Power Series Solutions of First Order Autonomous Algebraic Ordinary Differential Equations	Sebastian Falkensteiner
18:00 - 18:30	Computation of differential Chow forms for ordinary prime differential ideals	Wei Li

18:30 - 19:00	Group Classification of ODEs: a Challenge to Differential Algebra?	Dmitry Lyakhov
19:00 - 19:30	Power series solutions of systems of nonlinear	Daniel Robertz

S 07 Algebraic and Algorithmic Aspects of Differential and Integral Operato

Aula 9	title	speaker
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Tuesday 19th

15:30 - 16:00	The Jacobian algebras, their ideals and automorphisms	Vladimir Bavula
16:00 - 16:30	On the Parameter Estimation Problem for Integro-Differential Models	François Boulrier
16:30 - 17:00	Parametric b-functions for some hypergeometric ideals	Francisco-Jesus Castro-Jimenez
<i>coffee break</i>		
17:30 - 18:00	Reduction operators and completion of linear rewriting systems	Cyrille Chenavier
18:00 - 18:30	Desingularization in the q-Weyl algebra	Yi Zhang
18:30 - 19:00	Solution of non-homogenous Ordinary Differential Equations using Parametric Integral Method	David G. Zeitoun

Wednesday 20th

10:00 - 10:30	Low-Order Recombinations of C-finite Sequences	Maximilian Jaroschek
10:30 - 11:00	Some Properties and Invariants of Multivariate Difference-Differential Dimension Polynomials	Alexander Levin
<i>coffee break</i>		
11:30 - 12:00	Computer algebra and the Lanczos problems in arbitrary dimensions	Jean-François Pommaret
12:00 - 12:30	Symbolic computation for integro-differential-time-delay operators with matrix coefficients	Jamal Hossein Poor
12:30 - 13:00	Algebraic proofs of operator identities	Clemens Raab
13:00 - 13:30	Definite Integration of D-finite Functions via Generalized Hermite Reduction	Bruno Salvy

Thursday 21st

10:00 - 10:30	Effective criterion to test differential transcendence of special functions	Thomas Dreyfus
10:30 - 11:00	Observability and orders of derivatives of data	Sette Diop
<i>coffee break</i>		

S 08 Dynamic Geometry and Mathematics Education

Aula 10	title	speaker
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Wednesday 20th

10:00 - 10:30	A new approach to automated study of isoptic curves	Thierry Dana-Picard
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10:30 - 11:00	Exploration of dual curves using dynamic geometry and computer algebra system	Roman Hašek
<i>coffee break</i>		
11:30 - 12:00	Programming in KeTCindy with Combined Use of Cinderella and Maxima	Setsuo Takato
12:00 - 12:30	Discovering properties of bar linkage mechanisms based on partial Latin squares by means of Dynamic Geometry Systems	Raúl M. Falcón
12:30 - 13:00	Issues and challenges about instrumental proof	Philippe R. Richard
13:00 - 13:30	Dynamic Geometry and Computer Algebra Systems in Mathematics instruction	Round table

S 09 Computer Algebra in Coding Theory and Cryptography

Aula 7	title	speaker
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Tuesday 19th

10:30 - 11:00	On varieties and codes defined by quadratic equations	Ruud Pellikaan
<i>coffee break</i>		
11:30 - 12:00	Cyclic structures in convolutional codes and free distance	F. Javier Lobillo
12:00 - 12:30	On additive cyclic codes over chain rings	Edgar Martínez-Moro
12:30 - 13:00	Computer algebra tales on Goppa codes and McEliece cryptography	Narcís Sayols
13:00 - 13:30	Satisfiability modulo theory in finding the distance distribution of binary constrained arrays	Putranto Utomo
lunch		
15:30 - 16:00	Binary Isodual Codes Having an Automorphism of Odd Prime Order	Stefka Bouyuklieva
16:00 - 16:30	Quantum codes from constacyclic codes over the finite ring $F_p + uF_p + vF_p$	Abdullah Dertli
16:30 - 17:00	Constacyclic and Cyclic Codes over the Class of Finite Rings $F_{2^k} + uF_{2^k} + u^2F_{2^k} + vF_{2^k}$	G.Gözde Güzel
<i>coffee break</i>		
17:30 - 18:00	On the skew cyclic codes and the reversibility problem for DNA 4-bases	Yasemin Çengellenmis
18:00 - 18:30	The enumeration of Hermitian self-dual cyclic codes over finite chain rings	Arunwan Boripan

Wednesday 20th

10:00 - 10:30	Multiplying Dimension in Abelian Codes	Diana H. Bueno-Carreño
10:30 - 11:00	Self-dual codes over chain rings	Simon Eisenbarth
<i>coffee break</i>		
11:30 - 12:00	On the rank and kernel of new HFP-codes	Emilio Suárez-Canedo
12:00 - 12:30	Generalized HammingWeights of Binary Linear Codes	Irene Márquez-Corbella

S 10 Parametric Polynomial Systems

Aula 10	title	speaker
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Thursday 21st

16:30 - 17:00	Presentation of "The Gröbner Cover"	Antonio Montes
<i>coffee break</i>		
17:30 - 18:00	A canonical representation of continuity of the roots of a parametric zero dimensional multi-variate polynomial ideal	Yosuke Sato
18:00 - 18:30	Fitting a Sphere to Point Cloud Data via Computer Algebra	Robert H. Lewis
18:30 - 19:00	Resultants, Implicit Parameterizations, and Intersections of Surfaces	Robert H. Lewis
19:00 - 10:30	An overview on marked bases and applications	Cristina Bertone

Friday 22nd

10:30 - 11:00	Computation methods of b-functions associated with μ -constant deformations —Case of inner modality 2—	Katsusuke Nabeshima
<i>coffee break</i>		
11:30 - 12:00	An effective method for computing Grothendieck point residues	Shinichi Tajima
12:00 - 12:30	An algorithm for computing Grothendieck local residues II —general case—	Katsuyoshi Ohara

S 11 Algorithms for Zero-Dimensional Ideals

Aula 1	title	speaker
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Tuesday 19th

10:30 - 11:00	Fast Gröbner basis computation and polynomial reduction in the generic bivariate case	Robin Larrieu
<i>coffee break</i>		
11:30 - 12:30	Special Properties of Zero-Dimensional Ideals: new Algorithms	Lorenzo Robbiano
12:30 - 13:00	Computing Subschemes of the Border Basis Scheme	Martin Kreuzer
13:00 - 13:30	On the computation of algebraic relations of bivariate polynomials	Simone Naldi
15:30 - 16:30	On the decoding of interleaved and folded Reed-Solomon codes	Daniel Augot
16:30 - 17:00	Solving and bonding 0-dimensional ideals: Möller Algorithm and Macaulay Bases	Teo Mora
<i>coffee break</i>		
17:30 - 18:00	Combinatorics of ideals of points: a Cerlienco-Mureddu-like approach for an iterative lex game	Michela Ceria

18:00 - 18:30	Computing Recurrence Relations of n -dimensional Sequences Using Dual of Ideals	Hamid Rahkooy
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Wednesday 20th

10:00 - 11:00	Border basis, Hilbert Scheme of points and flat deformations	Mariemi Alonso
<i>coffee break</i>		
11:30 - 12:00	De Nugis Groebnerialium 5: Noether, Macaulay, Jordan	Teo Mora
12:00 - 12:30	Signature-based Criteria for Möller's Algorithm for Computing Gröbner Bases over PIDs	Thibaut Verron
12:30 - 13:00	Computing and using minimal polynomials	Anna M. Bigatti

S 12 Numerical Differential and Polynomial Algebra

Aula 9	title	speaker
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Friday 22nd

<i>coffee break</i>		
11:30 - 12:00	Symbolic-numeric methods for simulation of cosserat rods	Dmitry Lyakhov
12:00 - 12:30	A symbolic-numeric method to determine symmetry of approximate differential equations	Zahra Mohammadi
12:30 - 13:00	Challenges in Numerical Differential Algebra	Greg Reid

Sponsor Session

Aula Magna	speaker
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Thursday 21st

15:30 - 16:30	New features in Maple 2018	Daniel Skoog
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Business meeting

Aula Magna	
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Thursday 21st

13:00 - 13:30		
13:30 - 13:45	Presentation of ACA 2019 Montréal	Michel Beaudin